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CIA - 1

Mobile Computing.

1) Given data

Area = 1200km²

Cell radius = 1.8km

Spectrum = 48MHz

Bandwidth = 200KHz.

Cluster sizes N=7 and N=4.

Cell reduction = 15%

Area :-

$$\frac{3\sqrt{3}}{2}(R^2) = \frac{3\sqrt{3}}{2}(1.8)^2 = 8.4\text{km}^2$$

$$= 2.5981 \times 3.2400$$

Total cells

$$\frac{1200}{8.42} = 143//$$

Channel Total value

$$\frac{48000}{200} = 240$$

$$= 240$$

cluster sizes	N = 7	N = 4
Total channels	240	240
	240/7	240/4
	= 34.2	= 60
Reuse ratio	4.56	3.6

Formula

$$D = R\sqrt{3N} \qquad Q = \sqrt{3N}$$

∴ N = 7 satisfies and more suitable configuration for cellular network performance

Reuse 4.5 ration than N = 4.

2) Given data

15 = handoffs

2 = roaming transitions

3 = failed handoffs.

Normal handoff delay :-

$$15(120+80) = 3000ms$$

$$3000ms = 3S$$

Reconfig delay :-

$$3 \times 400 = 1200ms$$

$$1200ms = 1.2S$$

Reaming delay :-

$$2(350+250) = 2(600)$$

$$1200ms$$

$$= 1.2S$$

Total interruption delay

$$= 3+1.2+1.2$$

$$= 3+2.4$$

$$= 5.4$$

Data potential delay:-

$$512 \times 5.4 = 2764.8 \text{ //}$$

$$2764.8/8 = 345.6 \text{ //}$$

∴ The system fails the interruption of limit of 3sec cause interruption delay is 5.4s. which is greater than 3sec.

5) Given

2000 - mobile devices

Gsm provides 9.6kbps.

throughput - 64Kbps

10% - additional transmission.

5G provides 25 mbps

Calculate data volume

$$= \frac{120 \times 1000}{2000}$$

$$= 60MB$$

Into Bits

60MB X 8

= 480mb

Individual transfer time :-

$$= \frac{2000}{480}$$

$$= 4.166 s$$

Convert it into hours

= 13.87 hours

IN SG1

$$= \frac{25}{10}$$

$$= 2.5$$

$$= 60$$

IN GPRS

$$= \frac{64}{10}$$

$$= 1.6$$

Calculate the percentange of reduct :-

$$= \frac{T_{GPRS} - T_{SG1}}{T_{GPRS}}$$

$$= \frac{1600000 - 60}{1600000} = 9.9999\%$$

Given data

200 - medical Imaging files.

size - 18MB.

data - 20kbps

Capacity - 18%

Throughput - 85%

12% - packet retransmission.

T = 2.5 seconds.

Percentage reduction :-

$$85 \times 12$$
$$= 10200.000$$

time achieved by UMTS

$$2.5 \text{ sec}$$
$$2.5 \times 60 = 150.$$

size

$$18 \times 8 = 144.0\%$$

Capacity - 18%

3) (i) Given data

12 - BaseTS, 3 - sectors, 4 - transceivers.

Timeslots - 8

channel utilization - 80 %

minimum no. of TRXs.

$$\frac{12}{3 \times 4} = \frac{12}{12} = 1 \times 8$$

8 timeslots

$$\frac{1200}{1.2} \times 60 = 1000 \times 60 = 6000$$

E1 Abis link:-

$$\frac{2,048 \text{ mbps} \times 16 \text{ kbps}}{1000}$$

$$= \frac{2048 \times 16}{1000}$$

$$= \frac{32768}{1000}$$

$$= 32.7680$$

(ii) FDMA uses 30KHz channels with frequency N = 7

TDMA uses 200KHz carries with 8 users N=4

CDMA uses 1.25MHz with chip rate 1.2288 Mcps.

Datarate = 9.6Kbps

Eb/No of 8

voice activity factor = 0.4

reuse efficiency = 0.6.

FDMA :-

$$\frac{30 \times 1000}{7} = \frac{30000}{7} = 4285.7$$

TDMA :-

$200 \times 8 = 1600$

$$\frac{1600}{N=4} \qquad \frac{400}{\frac{1600}{4}} = 400$$

CDMA :-

1.25MHz chip rate of 1.2288 mcps.

Datarate = 9.6Kbps.

Eb/No : 8

9.6 × 8

76.800

Comparing FDMA, CDMA, TDMA

CDMA uses 1.25MHz

with chip rate 1.2288

1.25 × 1.2288

= 1.5350.